Anqi Tan

Ph.D. Student, Bioengineering · Seeking 2027 Internship in Optical & Hardware Engineering

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Designs portable optical instruments end to end: optics, mechanical design, embedded hardware, and the Android and analysis software that make them usable. Available May–Aug 2027 (on-site or remote) and part-time remote during the school year.

EDUCATION

Ph.D. in Bioengineering, University of Illinois Urbana-Champaign Aug 2022 – Present
Advisor: Prof. Brian T. Cunningham · Nanosensors Group

B.S. in Mechanical Engineering (dual degree), UIUC & Zhejiang University Aug 2018 – Jun 2022

EXPERIENCE

Graduate Research Assistant, Nanosensors Group, UIUC Sep 2022 – Present

- Designed and built a **~$100 pocket-size (66×68×95 mm) NIR fluorescence reader** for lateral flow strips end to end: 750 nm excitation and optical filtering, Raspberry Pi camera hardware, and an **Android app (Kotlin)** for acquisition and automated test/control-line quantification; imaging comparable to a benchtop ImageQuant scanner (first-author manuscript in preparation).
- Wrote the reader's analysis pipeline (**automated test/control-line detection, background correction, 4PL calibration**) that converts strip images into analyte concentrations.
- Lead reader development for a **$4.87M DoD (PRMRP)** point-of-care sepsis/ARDS project with Stanford, WashU, and UCSF.
- Lead the Live Hormone project: a salivary cortisol lateral flow test (**LOD ≤0.12 ng/mL**) with a phone-controlled reader, tested with **153 adults**; **oral presentation, APHA 2026**.

Research Assistant Intern, BrainCo (Zhejiang Qiangnao Technology), Hangzhou, China May 2022 – Aug 2022

- Supported two EEG studies at the company's brain-science institute: sleep assistance with acoustic and electrical stimulation, and fatigue monitoring with multimodal EEG.

Undergraduate Research Leader, ZJU-UIUC Institute 2019 – 2021

- Led a national-level undergraduate research project designing a re-localizable remote-center-of-motion mechanism for minimally invasive surgery (**first-author paper, IEEE ICARM 2021**).

SELECTED PROJECTS

Mimicry Stage Lighting Control System (Senior Design, mechanical lead) Feb 2022 – May 2022

- Designed a two-axis light manipulator (MATLAB kinematics, Euler–Lagrange torque analysis) and a 3D-printed/laser-cut portable structure (9.3 kg, 4.5 h battery); the light follows a handheld ESP32 + IMU controller in real time.

PUBLICATIONS

First author: IEEE ICARM 2021 · pocket fluorescence reader manuscript (in preparation). **Co-author:** 4 journal papers (*Chemical Reviews*, *Lab on a Chip*, *ACS Nano Medicine*, *Biosensors*) and 1 Elsevier book chapter. **Talks:** APHA 2026 (oral), LOAC 2026.

HONORS & TEACHING

1st Place, Ashby Prize Computational Science Hackathon, NCSA (2025) · National Silver Medal, China "Internet+" Innovation & Entrepreneurship Competition (2021) · TA, BIOE/ECE 415 Biomedical Instrumentation Lab (2025) and ME 170 Computer-Aided Design (3 semesters)

SKILLS

Optics & Hardware: fluorescence imaging optics (NIR excitation, filter selection), camera modules, Raspberry Pi, ESP32, Arduino, LabVIEW, IMU sensors
Engineering: SolidWorks, Creo, Fusion 360, 3D printing, laser cutting, optical/instrument design, COMSOL, FEA, design for assembly
Software: Python, MATLAB, Android app development (Kotlin), image processing, curve fitting (4PL)
Languages: English (fluent), Chinese (native)